Conversational AI & Chatbots: Building Smart Assistants

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Introduction to AI

"Every once in a while, a new technology, an old problem, and a big idea turn into an innovation."

— Dean Kamen

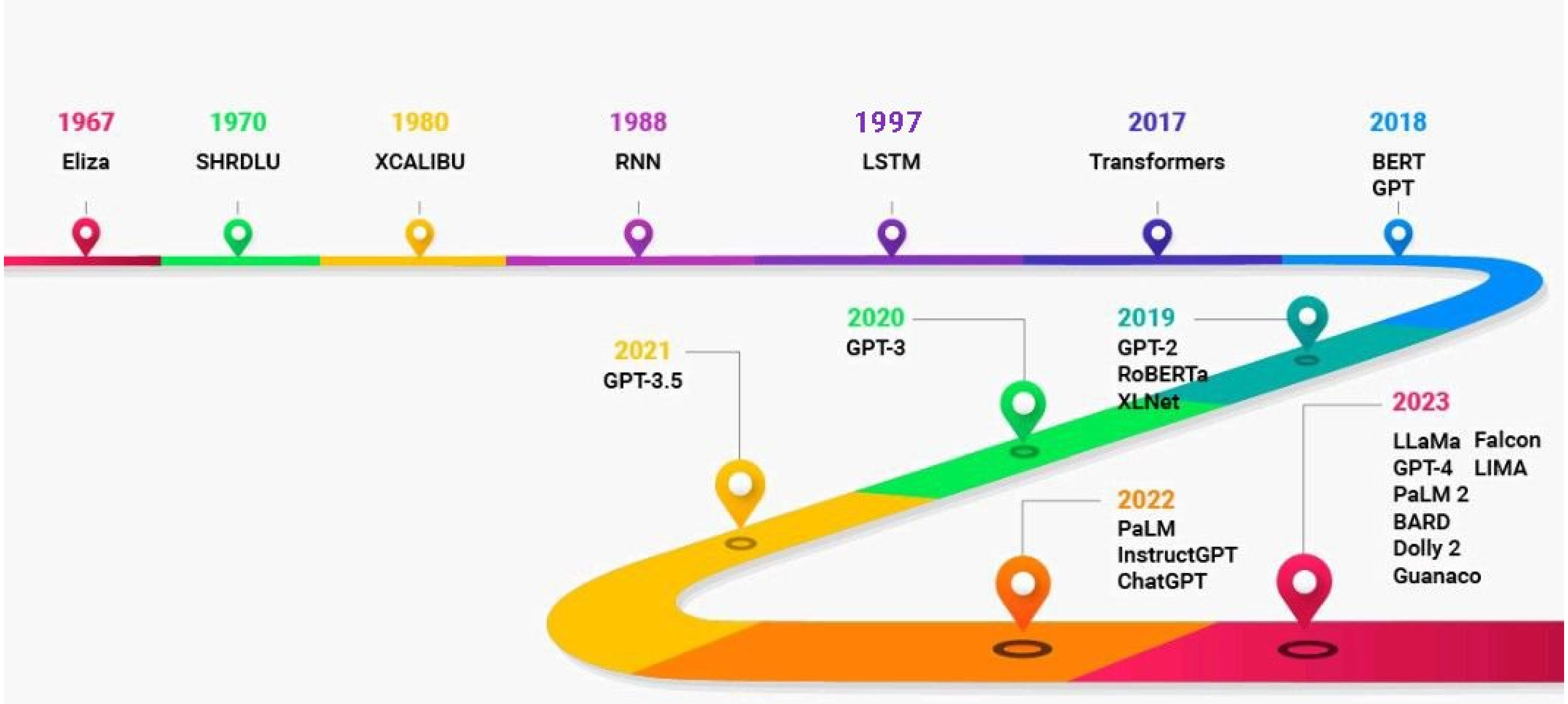
What is AI?

Artificial Intelligence (AI) is the simulation of human intelligence in machines designed to perform tasks that typically require human cognition. These tasks include learning from data, reasoning, problem-solving, understanding language, and perceiving the environment.

Evolution of AI: From Rule-Based Systems to Deep Learning

Rule-Based Systems	AI Systems
Predefined *if-then* rules	Data-driven learning and reasoning
Deterministic (fixed outcomes)	Probabilistic (context-based outcomes)
Simple, structured tasks	Complex, unstructured tasks
Cannot learn or adapt	Learns and improves from data
Static; requires manual updates	Dynamic; self-updating and adaptive
Predictable, rule-driven	Changing, data-rich environments
Tax calculators, basic chatbots	ChatGPT, recommendation system, self-driving cars

Evolution of AI: From Rule-Based Systems to Deep Learning



Introduction to Large Language Models (LLMs)

What is LLM?
A Large Language Model (LLM) is an advanced artificial intelligence model designed to understand, generate, and process human language. Trained on vast amounts of text data, LLMs use deep learning techniques, especially transformer architectures, to perform tasks like text generation, translation, summarization, and question answering.

Few Popular LLMs

GPT - 4

Gemini

Claude

deepseek

YandexGPT

How LLMs Work?

What is Tokenization?
Tokenization is the process of breaking down text into smaller units called tokens, which can be words, subwords, or characters. In natural language processing (NLP), tokenization helps AI models understand and process language by converting text into a structured format.

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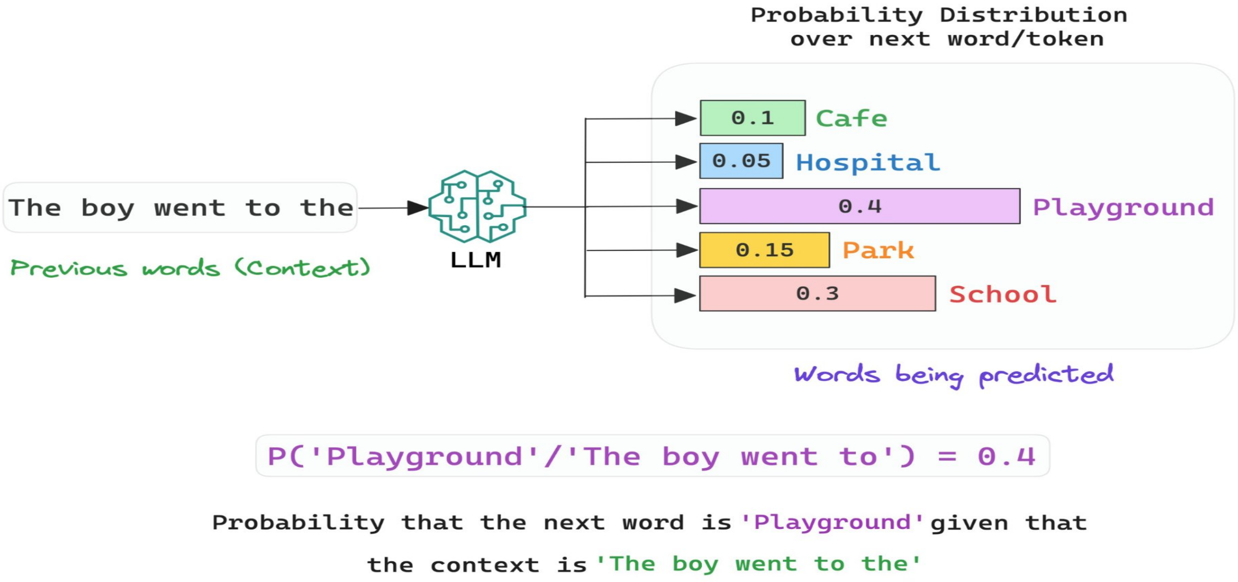
How LLMs Work?

Text generation occurs **token by token**, where a LLM predicts the probability of the **next token** from all tokens in its **vocabulary**. The token with the **highest probability** is selected as the **next generated token** in the sequence.



How LLMs are Trained?

1. **Pre-Training** → Learn general language understanding.

1. **Supervised Fine-Tuning** → Adapt to specific tasks.

1. **RLHF** → Align with human preferences.

1. **Post-Training Refinements** → Optimize performance.

1. **Continuous Learning & Deployment** → Maintain and improve the model.

How LLMs are Built: Infrastructure Requirements Insight

Llama 1

Pre-training data corpus: ~ 1.4 Trillion Tokens

LLM Model Size: ~ 65 Billion Parameter

Number of GPU: ~ 2048 A100 (80GB VRAM) for 21 days

How LLMs are Built: Infrastructure Requirements Insight

Llama 2

Pre-training data corpus: ~ 2 Trillion Tokens ~ 5.8 TeraByte

LLM Model Size: ~ 70 Billion Parameter

Number of GPU Hours: ~ 1,720,320 GPU hours

How LLMs are Built: Infrastructure Requirements Insight

Llama 3

Pre-training data corpus: ~ 15 Trillion Tokens ~ 44 TeraByte

LLM Model Size: ~ 405 Billion Parameter

Number of GPU: ~ 16000 H100 (80GB VRAM)

Computational Power Needed for Inference

$$M = ((P \times 4B)/(32/Q)) \times 1.2$$

Where,

- **M** = Required GPU memory in **gigabytes (GB)**
- **P** = Number of **parameters** in the model
- **4B** = **4 bytes** per parameter (since most models use FP32 or FP16 precision)
- **Q** = Number of **bits** per parameter when loading the model (e.g., 16-bit for FP16, 8-bit for quantized models)
- **1.2** = A 20% **overhead** factor

Computational Power Needed for Inference

$$M = ((P \times 4B)/(32/Q)) \times 1.2$$

How Much Memory Does a 7B Parameter Model Need for 16-bit precision?

M = ((7,000,000,000 × 4)/(32/16)) × 1.2 M

= ((28,000,000,000)/2) × 1.2 M

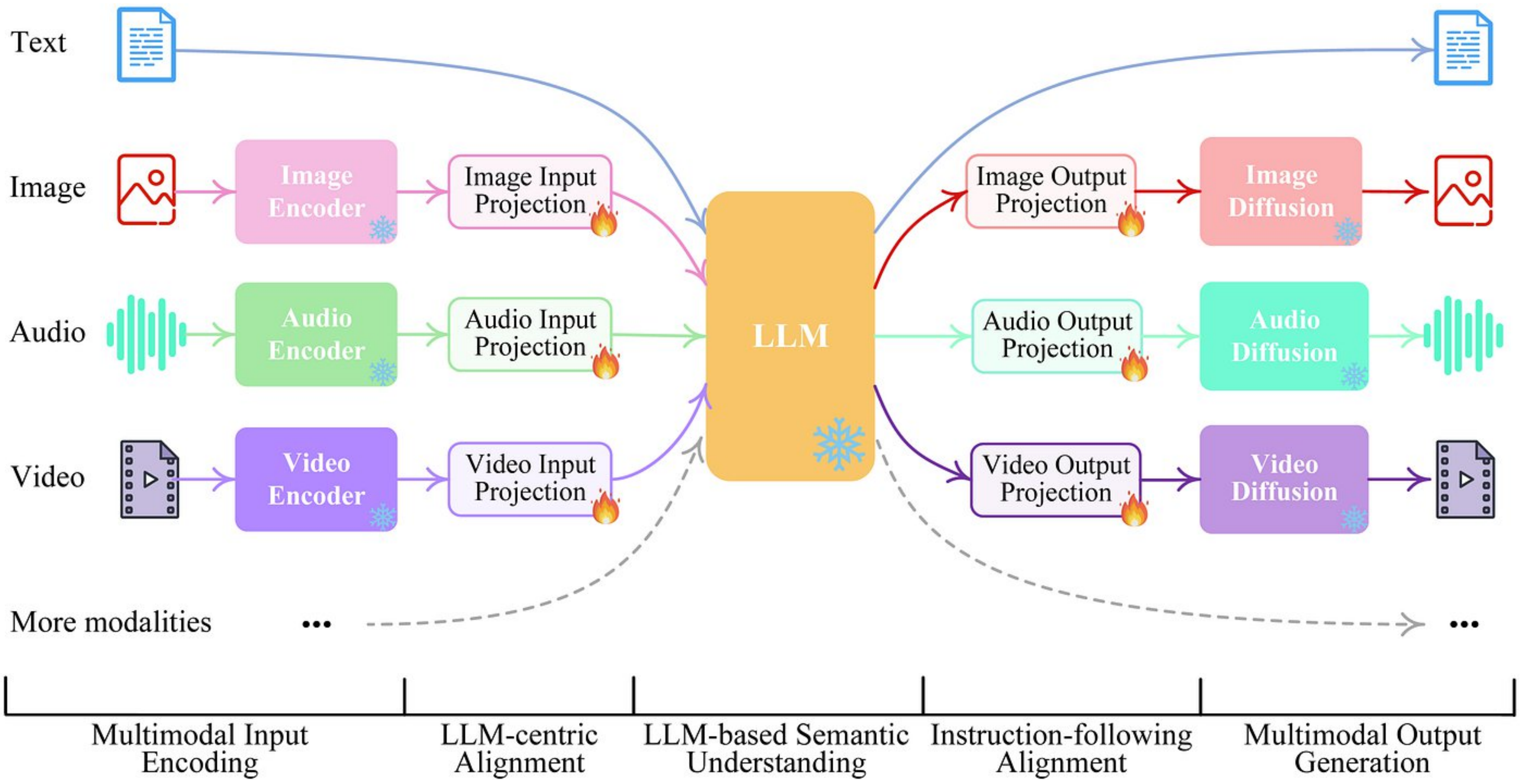
= 14,000,000,000 × 1.2 M

= 16.8 GB

Popular Open-Source LLMs

- LLaMA (Meta)
- BLOOM
- Falcon (UAE)
- Mistral
- OPT
- XGen
- Vicuna
- YaLM
- GPT-NeoX and GPT-J
- Pythia (Apple)
- Dolly
- MPT
- Zephyr
- Qwen
- Gemma
- Mixtral

Multi-Modal AI: Combining Text, Images, and Audio



Use Cases of Multi-Modal LLMs in Real-World Applications

- **AI-powered Virtual Assistants** – Processing text, images, and voice for smarter interactions.
- **Medical Diagnosis & Assistance** – Analyzing reports, medical images, and patient queries.
- **Autonomous Vehicles** – Understanding road signs, objects, and spoken instructions.
- **Retail & Ecommerce** – Visual search, product recommendations, and chatbot support.
- **Customer Support Automation** – Handling text, voice, and image-based queries.

Where are we with Bangla?

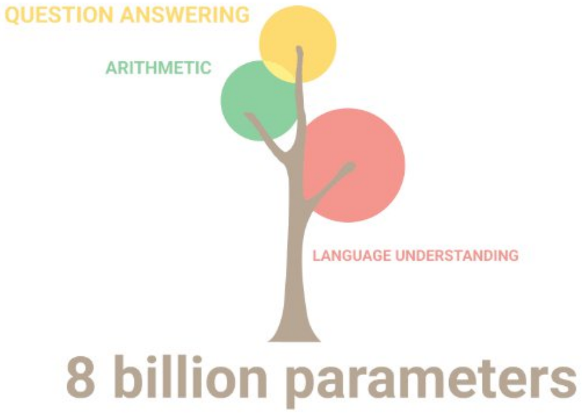
Why ChatGPT is not so good at Bangla?

- No Bangla specific Tokenizer
- Limited portion of Bangla samples in Pre-Training Dataset

Why we are not able to build BanglaGPT?

- No GPU data centers
- Limitation of Pre-Training Dataset

Is Limited Data & Training Beneficial?



Approaches to Improve Bengali Language Processing in AI

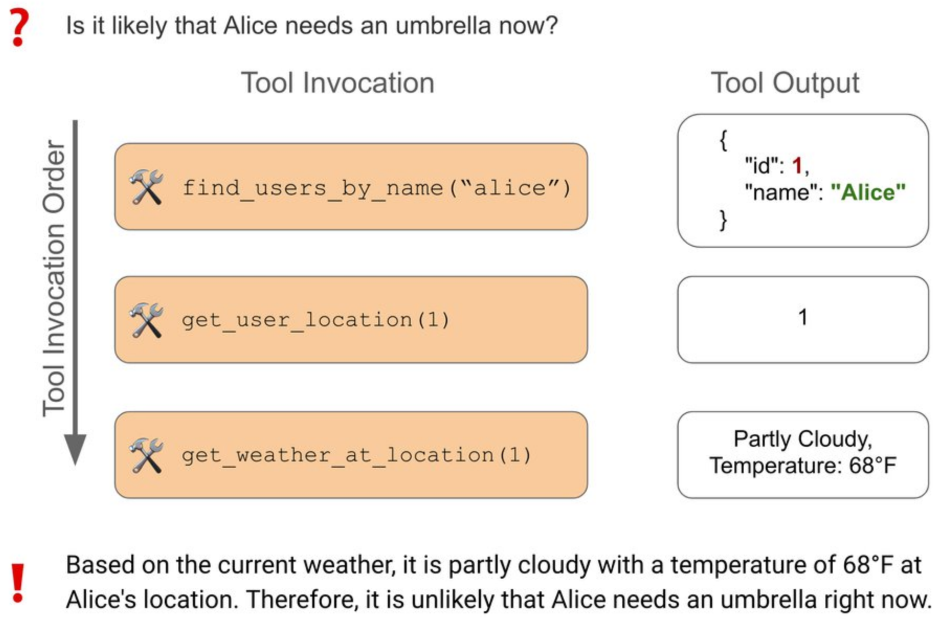
Challenges for Limited Training

- Low language adaptability
- Low reasoning capability
- Repetitive token generation issue

Easiest Solution

Tools

Functions or APIs that an LLM can call to get external information or perform specific tasks.

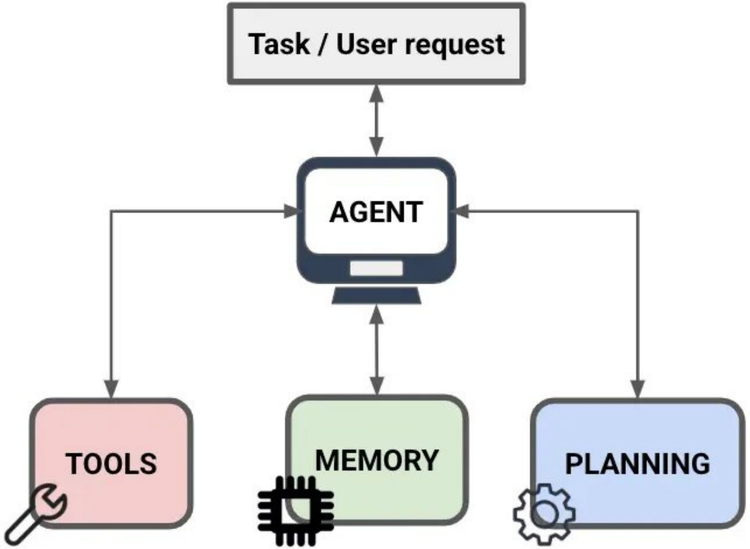


Tools

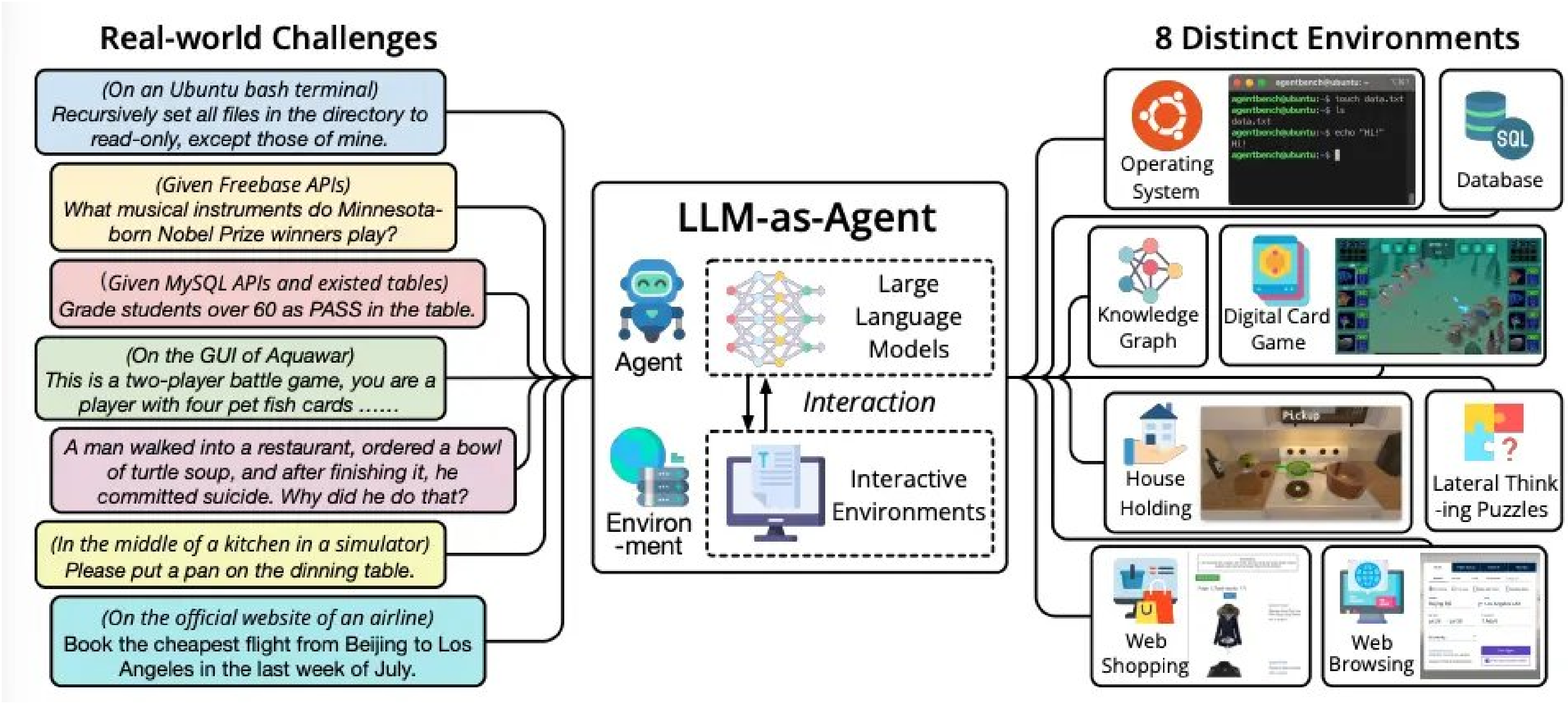
- Web Search & Knowledge Retrieval
- Database Querying & Vector Search
- Math & Code Execution
- API Call & External Services
- File Handling & Document Processing
- Image & Video Processing
- Text-to-Speech (TTS) & Speech Recognition
- Task Automation & Workflow Execution
- Email & Communication
- Security & Content Moderation

Agents

Autonomous systems that plan, reason, and execute actions using tools dynamically.



Agents



Real-world Challenges
(On an Ubuntu bash terminal) Recursively set all files in the directory to read-only, except those of mine.
(Given Freebase APIs) What musical instruments do Minnesota-born Nobel Prize winners play?
(Given MySQL APIs and existed tables) Grade students over 60 as PASS in the table.
(On the GUI of Aquawar) This is a two-player battle game, you are a player with four pet fish cards ......
A man walked into a restaurant, ordered a bowl of turtle soup, and after finishing it, he committed suicide. Why did he do that?
(In the middle of a kitchen in a simulator) Please put a pan on the dinning table.
(On the official website of an airline) Book the cheapest flight from Beijing to Los Angeles in the last week of July.
LLM-as-Agent
Agent
Large Language Models
Interaction
Environ -ment
Interactive Environments
8 Distinct Environments
Operating System
Database
SQL
Knowledge Graph
Digital Card Game
House Holding
Lateral Think -ing Puzzles
Web Shopping
Web Browsing

RAG System

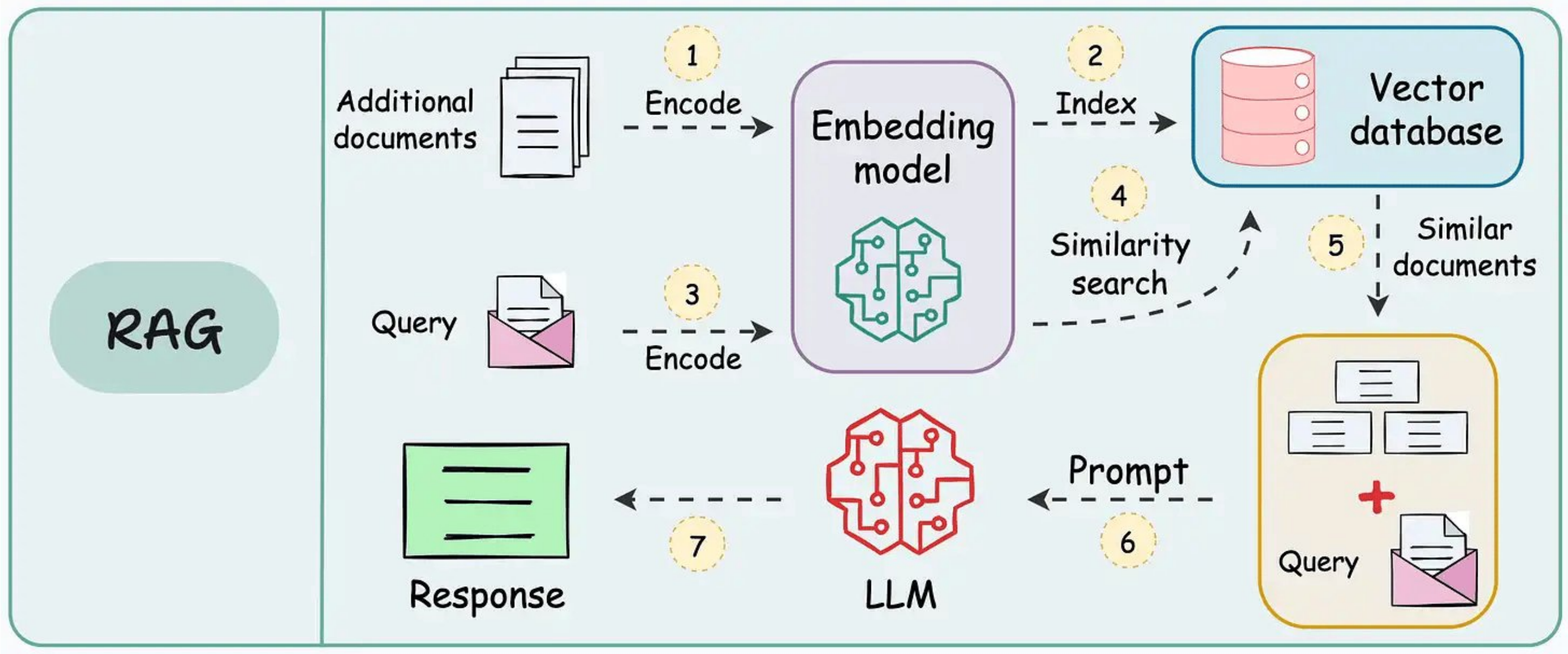
What is RAG System?

A Retrieval-Augmented Generation (RAG) system combines information retrieval with text generation. It first retrieves relevant documents from a knowledge base and then uses an LLM to generate responses based on both the retrieved data and the input query.

Why RAG is Important?

- **Enhances Accuracy** – Uses real data to generate reliable responses.
- **Reduces Hallucinations** – Limits AI-generated misinformation.
- **Improves Context Awareness** – Retrieves relevant information dynamically.
- **Enables Real-time Updates** – Uses the latest knowledge sources.
- **Supports Domain-Specific Knowledge** – Integrates specialized datasets.
- **Optimizes Token Usage** – Reduces reliance on model memory.
- **Boosts Explainability** – Provides source references for responses.

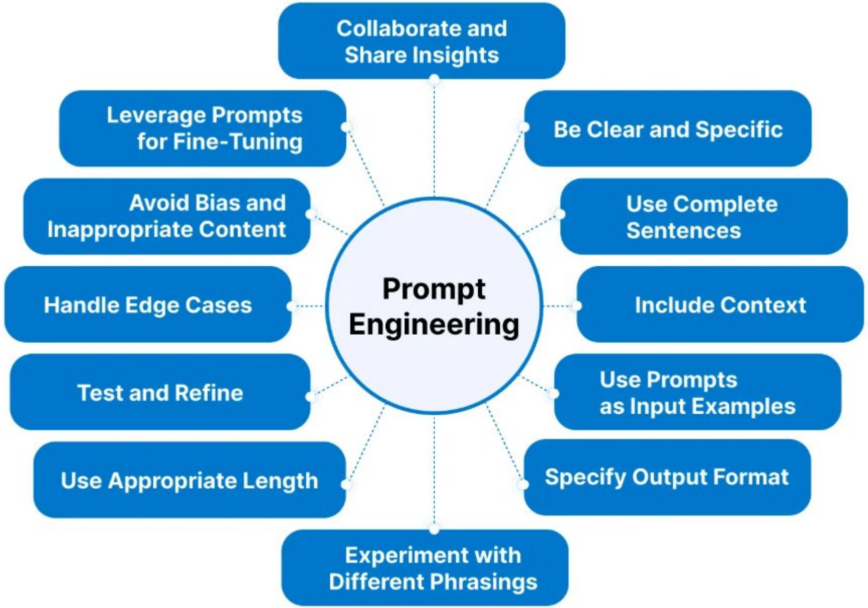
RAG System Pipeline



Prompt Engineering

What is Prompt Engineering?

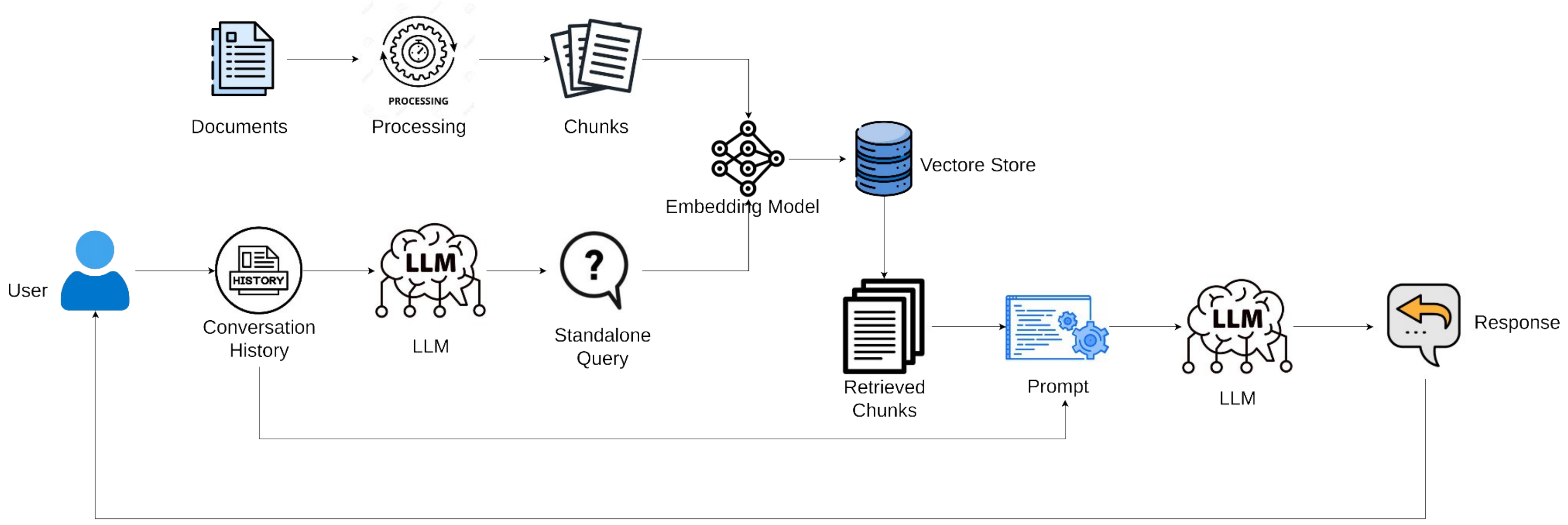
Prompt engineering is the process of designing effective inputs for AI models to get better responses. It involves crafting clear, structured, and optimized prompts to improve accuracy and relevance.



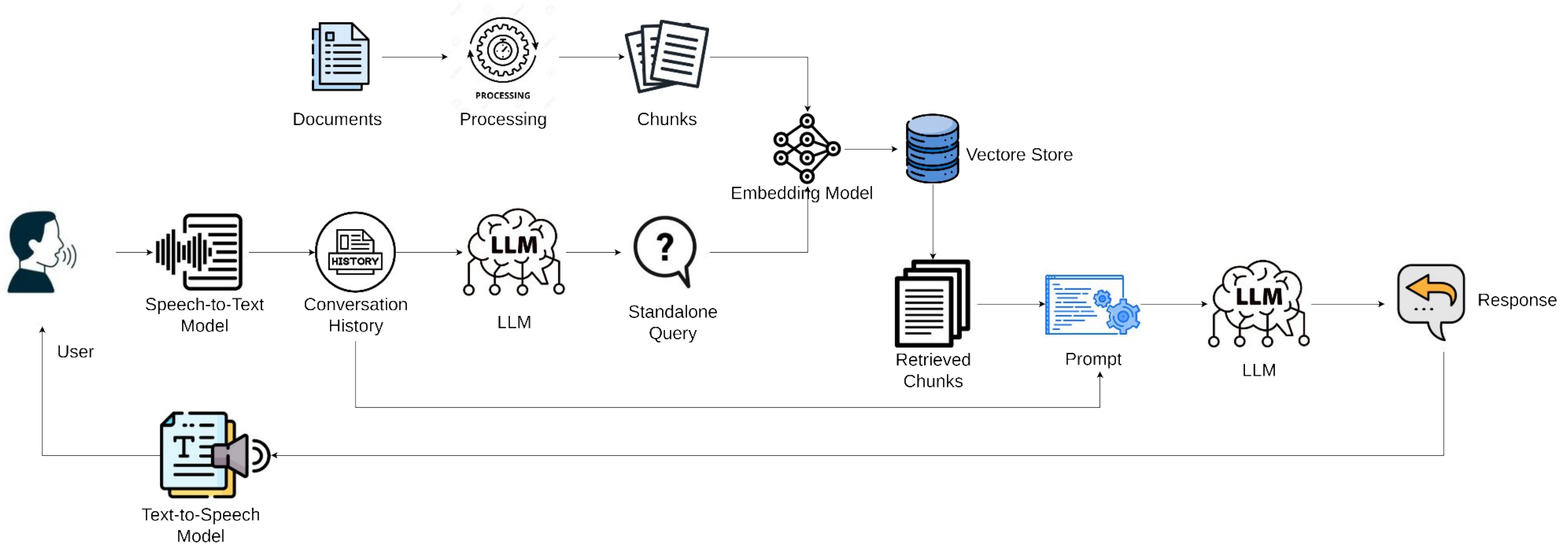
Case Study: Conversational AI & Chatbots

- Customer Support Automation
- E-commerce Assistance
- Healthcare Chatbots
- Legal Consultation Bots
- Banking & Finance Support
- Enterprise Knowledge Assistants
- Educational Chatbots

Case Study: Conversational AI & Chatbots



Case Study: Voice Agents



RAG System Demonstration

Announcement: RAG System Project Showcase

- Deadline: 15 April, 2025
- Resource: https://rb.gy/655rc7

Top 2 will be awarded!

Do you have any question?

Feedback:
https://forms.gle/uz21fWwnCsvcKXpG8

Thank You